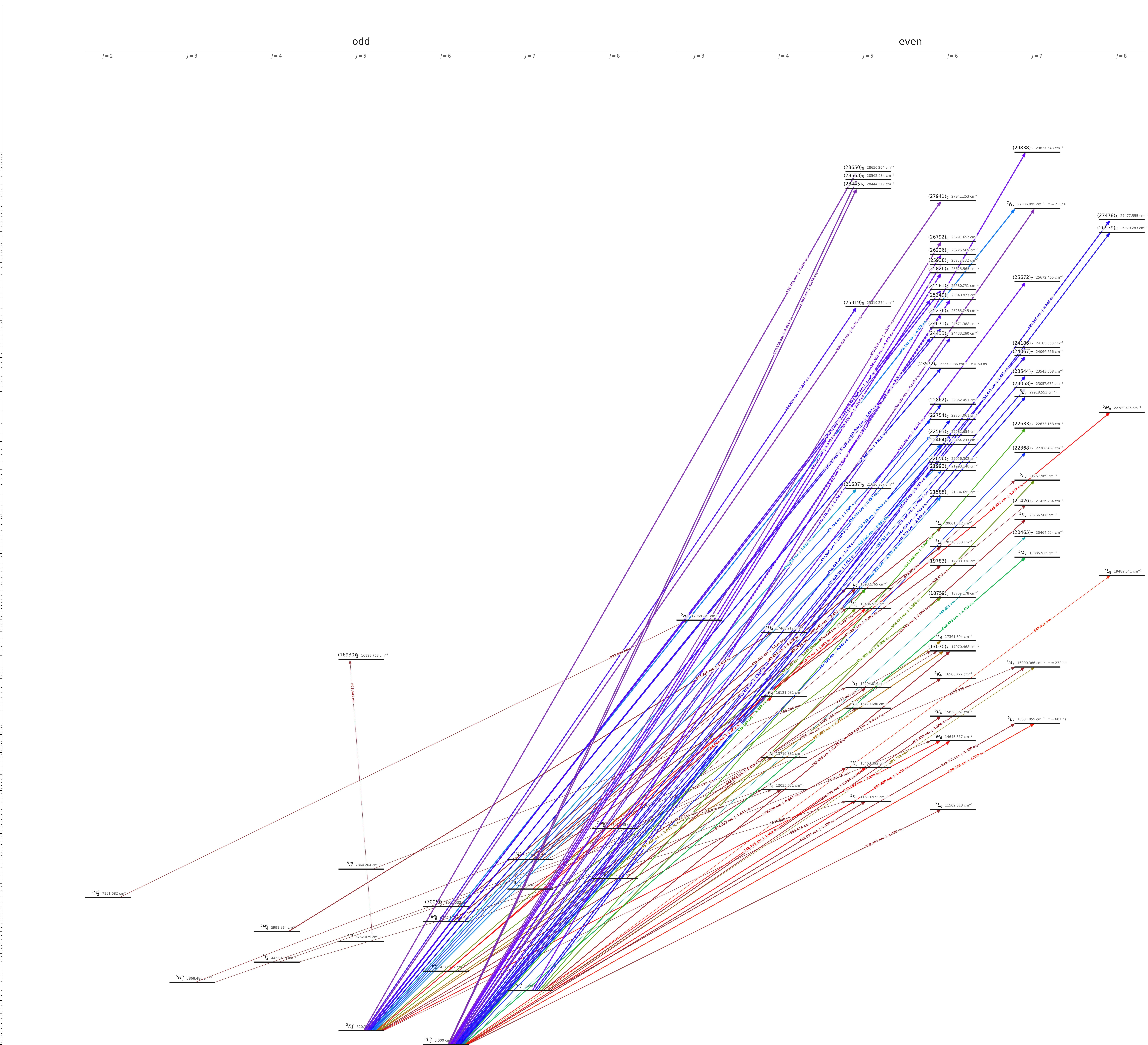


U<sup>+</sup> ionisation limit    49958.40 cm<sup>-1</sup> = 6.19405 eV (200.167 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|V|e|f||\epsilon_0|$  in  $\text{ea}_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory, = model or semi-empirical calculation.  
 Arrow width grows with the Einstein coefficient  $A$ . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).



| Transition              | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ U(1J) $ (ea) | A (s <sup>-1</sup> ) | $f/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|-------------------------|--------------------|--------------------|-------------|----------------|----------------------|--------------------|------------|--------------------------------------------------------------|
| $^1L_6 - (^2S_{3/2})_0$ | 396.2633           | 396.1512           | 756.5486    | -              | -                    | -                  | -          | 396.263 nm                                                   |
| $^1L_8 - ^1M_5$         | 591.7025           | 591.5385           | 506.6608    | -              | -                    | 0.686              | -          | 591.54 nm: classic first-step line of uranium laser spect... |
| $^1L_8 - ^1L_5$         | 639.7193           | 639.4525           | 468.6312    | 1.369          | 9.670e+05            | 0.2622             | 58.6       | 639.542 nm                                                   |
| $^3K_5 - ^1L_5$         | 713.0865           | 712.8900           | 420.4153    | 1.258          | 6.800e+05            | -                  | -          | 713.087 nm                                                   |
| $^1M_6 - ^1L_5$         | 763.3846           | 763.7175           | 392.7148    | 1.104          | 3.700e+05            | 0.686              | 8.58       | 763.385 nm                                                   |
| $^1M_8 - ^1L_5$         | 788.4112           | 788.1994           | 380.2488    | 3.138          | 3.700e+06            | -                  | -          | 788.411 nm: line used for Doppler-free saturated-absorpt...  |
| $^1L_8 - ^1K_5$         | 861.0316           | 860.7952           | 348.1782    | 2.039          | 1.200e+06            | -                  | -          | strong ground-state absorption line                          |

A,  $\nu$ , from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $f/2\pi$ : natural linewidth or  $\Gamma$  of upper level,  $10^{12}$  Hz

235 U hyperfine structure ( $I = 7/2$ )

| Level    | A (MHz)  | B (MHz)                                                                                                           | Shift of each F level from the centre of gravity (MHz)                                                               |
|----------|----------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $6^1S_0$ | 4074.00  | F=2: +3303.57 F=7: +2197.25 F=9: +1190.70 F=11: +202.99 F=13: -620.41 F=15: -1107.53 F=17: -1059.96 F=19: -252.84 |                                                                                                                      |
| $8^1S_0$ | 68.34    | 40.10                                                                                                             | F=2: +1452.64 F=7: +1275.57 F=9: +1029.91 F=12: -974.29 F=13: +333.13 F=17: -112.36 F=19: -619.70 F=21: -1186.02     |
| $4^1D_2$ | 56.310   | 412.00                                                                                                            | F=7: +313.03 F=9: +2118.90 F=11: +1081.36 F=13: -1064.15 F=15: -678.50 F=17: -122.98 F=19: -1082.05 F=21: -351.99    |
| $5^1D_3$ | 59.130   | 447.00                                                                                                            | F=2: +1636.47 F=7: +1343.32 F=9: +996.69 F=11: +132.59 F=13: +168.12 F=17: -279.70 F=19: -719.91 F=21: -1129.68      |
| $7^1D_5$ | 67.800   | 2366.00                                                                                                           | F=2: -720.08 F=7: -886.07 F=9: -1007.31 F=11: -1014.66 F=13: -823.62 F=15: -334.33 F=17: +568.45 F=19: +2015.30      |
| $7^3P_0$ | 54.390   | 609.00                                                                                                            | F=2: +1547.77 F=7: +1280.60 F=9: +926.11 F=11: +122.59 F=13: +111.29 F=15: -301.58 F=17: -678.96 F=19: -989.94       |
| $5^3P_1$ | 63.150   | 1239.00                                                                                                           | F=7: +223.12 F=9: +1273.35 F=11: +1517.12 F=13: +555.33 F=15: -19.71 F=17: -548.22 F=19: -972.62 F=21: -1237.42      |
| $5^3P_2$ | -99(6)   | 956(24)                                                                                                           | F=2: +2484.44 F=7: +1569.85 F=9: +132.46 F=11: +12.48 F=13: -288.20 F=15: -966.51 F=17: -1486.45                     |
| $3^1D_2$ | 20.300   | 4950(120)                                                                                                         | F=2: +1476.18 F=7: +702.83 F=9: -98.63 F=11: -783.57 F=13: -1175.21 F=15: -1064.63 F=17: -210.79 F=19: +1659.53      |
| $1^1D_2$ | -84.720  | 4422.00                                                                                                           | F=2: +4024.34 F=7: +2848.51 F=9: +1597.85 F=13: +392.89 F=15: +632.99 F=17: -1323.61 F=19: -1501.94 F=21: -970.14    |
| $1^3P_0$ | -171.000 | 2685.00                                                                                                           | F=2: +5791.19 F=9: +4538.01 F=11: +3122.26 F=13: +1613.49 F=15: +493.89 F=17: -1341.70 F=19: -2585.79 F=21: -3518.25 |

Measured A constants: sources in the data file.

| Transition                   | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|------------------------------|--------------------|----------|-------------|
| $^4\text{Li}^-$ (28563),     | 350.108            | 235-238  | 10340.0     |
| $^4\text{Li}^-$ (288445),    | 351.562            | 235-238  | 8990.0      |
| $^4\text{Li}^- - \text{H}$ , | 358.590            | 235-238  | 17480.0     |
| $^6\text{Li}^-$ (27941),     | 366.020            | 235-238  | 4800.0      |
| $^6\text{Li}^-$ (26792),     | 373.250            | 235-238  | 5910.0      |
| $^6\text{Li}^-$ (26626),     | 381.307            | 235-238  | 6420.0      |
| $^6\text{Li}^-$ (29838),     | 384.072            | 235-238  | 13800.0     |
| $^6\text{Li}^-$ (29854),     | 385.552            | 235-238  | 8950.0      |
| $^6\text{Li}^-$ (25826),     | 387.213            | 235-238  | 8180.0      |
| $^6\text{Li}^-$ (25672),     | 389.522            | 235-238  | 10880.0     |
| $^6\text{Li}^-$ (25349),     | 394.493            | 235-238  | 8420.0      |
| $^6\text{Li}^-$ (25236),     | 396.263            | 235-238  | 12600.0     |
| $^6\text{Li}^-$ (25581),     | 400.634            | 235-238  | 4320.0      |
| $^6\text{Li}^-$ (25349),     | 404.389            | 235-238  | 8570.0      |
| $^6\text{Li}^-$ (25319),     | 404.875            | 235-238  | 13280.0     |
| $^6\text{Li}^-$ (24433),     | 405.578            | 235-238  | 5630.0      |
| $^6\text{Li}^-$ (24433),     | 409.214            | 235-238  | 10500.0     |
| $^6\text{Li}^-$ (24067),     | 419.782            | 235-238  | 12700.0     |
| $^6\text{Li}^-$ (24671),     | 415.782            | 235-238  | 11390.0     |
| $^6\text{Li}^-$ (24433),     | 419.940            | 235-238  | 5820.0      |
| $^6\text{Li}^-$ (27478),     | 422.356            | 235-238  | 10460.0     |
| $^6\text{Li}^-$ (23544),     | 427.435            | 235-238  | 8210.0      |

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$